

CPSC 304 Project Cover Page

Milestone #: 1

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Group Number: 80

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By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above.

In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia

Project Milestone 1 - Project Proposal, Conceptual Design, ER Diagram

Topic: Urban Bike Infrastructure Maintenance System

Description

The domain of the application

The domain of our application is urban bike infrastructure maintenance and operations. It concerns the real-world processes involved in operating and maintaining a city-wide shared bicycle system, including ensuring that bicycles deployed across urban stations remain safe, functional, and available for public use. This domain involves coordination between physical assets, maintenance staff, and operational policies to support reliable and efficient urban transportation.

Database Specification

This database models the main maintenance and operational aspects of the urban bike infrastructure system. It focuses on the real-world processes that monitor bicycles deployed across different stations to ensure each remains safe, functional, and available for all customers to use. Therefore, the database also supports issue reporting, maintenance activity management, and the staff responsible for inspection and repair to support the whole system.

To fit our application's domain, the database separates the big thing into three subaspects: bicycle condition, issue reporting, and maintenance activities. Each of them has a few entities and relationships; these subaspects also have relationships between each other to build the whole maintenance and operation system.

In the bicycle condition part, the database represents bicycles as central entities that include regular and e-bike types, and records their deployment status, service history, and current condition, which can help staff have a general view of a bicycle for periodic inspection and maintenance. And each bike needs to be associated with stations, which help the system easily navigate the bikes' locations and availability across the city, which enables the system to assign an inspector and technician who are responsible for the same base area to fix the problem and recover the availability quickly.

In the issue reporting part, the database models Issue as the central entity. This part mainly shows how the issues reported on bikes work. Each issue needs to have its specific IssueID, report time, mechanical problems, and condition scores. A customer can report an issue, and an inspector is responsible to check whether it needs to be fixed by the maintenance system.

Then, the maintenance activities section captures information on repair priorities, completion status, quotations, and detailed maintenance tasks, reflecting real-world workflows in which a single maintenance job may involve multiple repair steps. Components used in repairs are explicitly modelled to enable the system to track parts, materials, prices, and inventory levels.

In addition, the database includes staff entities with specialised roles (managers, technicians, and inspectors). So the system can indicate who performs inspections, who carries out repairs, and who manages maintenance operations. Customers are also modelled to support issue reporting and usage tracking within the shared bike network, which can help the system identify possible malicious causes of bicycle damage.

Overall, the database addresses the practical needs of urban bike infrastructure maintenance by integrating bike deployment, issue reporting, and maintenance workflows. For example, when a customer reports an issue with a bike at a station, the system can record the issue, assign an inspector to check whether repair is needed, assign maintenance staff to each of the divided maintenance tasks, track repair tasks and components used, and update the bike's status once maintenance is completed. This supports the safety, availability, and efficient operation of a city-wide bike-sharing system.

ER Diagram of the Application

Entities Description

Bike (IsA)

Primary Key:

- BikeID

Attributes:

- Brand
- LastServiceDate
- Status
- DeploymentDate

IsA Relationship (Total & Disjoint):

- E-Bike (with an extra attribute: BatteryLevel)
- RegularBike

Component

Primary Key:

- ComponentID

Attributes:

- Name
- Price
- Material
- StockQuantity

Customer

Primary Key:

- CustomerID

Attributes:

- Email
- Phone#
- Name
- Address
- RegisteredDate

Issue

Primary Key:

- IssueID

Attributes:

- ReportTime
- ConditionScore(e.g., ranges from 0 to 5)
- Result(need to be repaired/not need to be repaired),

Maintenance (Owner of Maintenance Task)

Primary Key:

- MaintenanceID

Attributes:

- PriorityLevel
- CompletionStatus
- RepairQuotation

Maintenance Task (Weak Entity, depends on Maintenance)

Partial Key:

- TaskID

Attributes:

- Duration
- StartTime
- EndTie

Staff (IsA)

Primary Key:

- StaffID

Attributes:

- Name
- RoleID
- AssignedArea
- Contact

IsA Relationship:

- Manager
- Technician
- Inspector

Station

Primary Key:

- StreetAddress
- PostalCode
- StationName

Attributes:

- NumberOfBike
- TotalCapacity
- InstallationDate

Entity Relationships Description

ConsistOf

Entities Involved:

- Component
- Bike

Description:

- Represents the association between a bike and the individual components that belong to it. Each component may belong to at most one bike, while a bike must consist of multiple components.

Cardinality:

- Bike → Component: One-to-Many

Constraint:

- Key constraint on Component
- Total participation constraint on Bike

LocatedAt

Attributes:

- StartTime
- EndTime

Entities Involved:

- Bike
- Station

Description:

- Represents the station at which a bike is currently located within the bike-sharing system. A bike may be located at at most one station at a time; however, bikes that are in use or under maintenance may not be located at any station.

Cardinality:

- Bike → Station: Many-to-One

Constraint:

- Key constraint on Bike

Need

Entities Involved:

- Component
- MaintenanceTask

Description:

- Represents the individual components required to perform a maintenance task. Each component may be used in at most one maintenance task, while a maintenance task may require multiple components.

Cardinality:

- MaintenanceTask → Component: One-to-Many

Constraint:

- Key constraint on Component

PartOf

Entities Involved:

- MaintenanceTask
- Maintenance

Description:

- Represents the inclusion of maintenance tasks within a maintenance job. Each maintenance task belongs to exactly one maintenance job, while a maintenance job may consist of multiple maintenance tasks.

Cardinality:

- MaintenanceTask → Maintenance: Many-to-One

Constraint:

- Total participation on MaintenanceTask (a maintenance task cannot exist without being part of a maintenance job)
- Key constraint on Maintenance

Experience

Entities Involved:

- Bike
- Issue

Description:

- Represents issues that occur on a bike during usage or inspection. A bike may experience multiple issues over time, while each issue is associated with exactly one bike.

Cardinality:

- Bike → Issue: One-to-Many

Constraint:

- Key constraint and total participation on Issue

Report

Attributes:

- Description

Entities Involved:

- Customer
- Issue

Description:

- Represents customers reporting issues encountered during bike usage. Multiple customers may report the same issue, and a customer may report multiple different issues.

Cardinality:

- Customer → Issue: Many-to-Many

Constraint:

- No key or total participation constraints apply

Fix

Entities Involved:

- Maintenance
- Issue

Description:

- Represents maintenance jobs that address reported issues. A maintenance job may fix multiple issues, while each issue is fixed by exactly one maintenance job.

Cardinality:

- Maintenance → Issue: One-to-Many

Constraint:

- Key constraint on Issue

Perform

Entities Involved:

- Technician
- MaintenanceTask

Description:

- Represents technicians carrying out maintenance tasks. A technician may perform multiple maintenance tasks, while each maintenance task is performed by exactly one technician.

Cardinality:

- Technician → MaintenanceTask: One-to-Many

Constraint:

- Key constraint on MaintenanceTask

Supervise

Entities Involved:

- Manager
- Staff

Description:

- Represents a manager supervising staff members within the organization. A manager may manage multiple staff members, while each staff member is managed by exactly one manager.

Cardinality:

- Manager → Staff: One-to-Many

Constraint:

- Key constraint and total participation constraint on Staff

Check

Entities Involved:

- Inspector
- Issue

Description:

- Represents an inspector examining an issue to assess its condition or determine whether maintenance is required. An inspector may check multiple issues, while each issue is checked by at least one inspector.

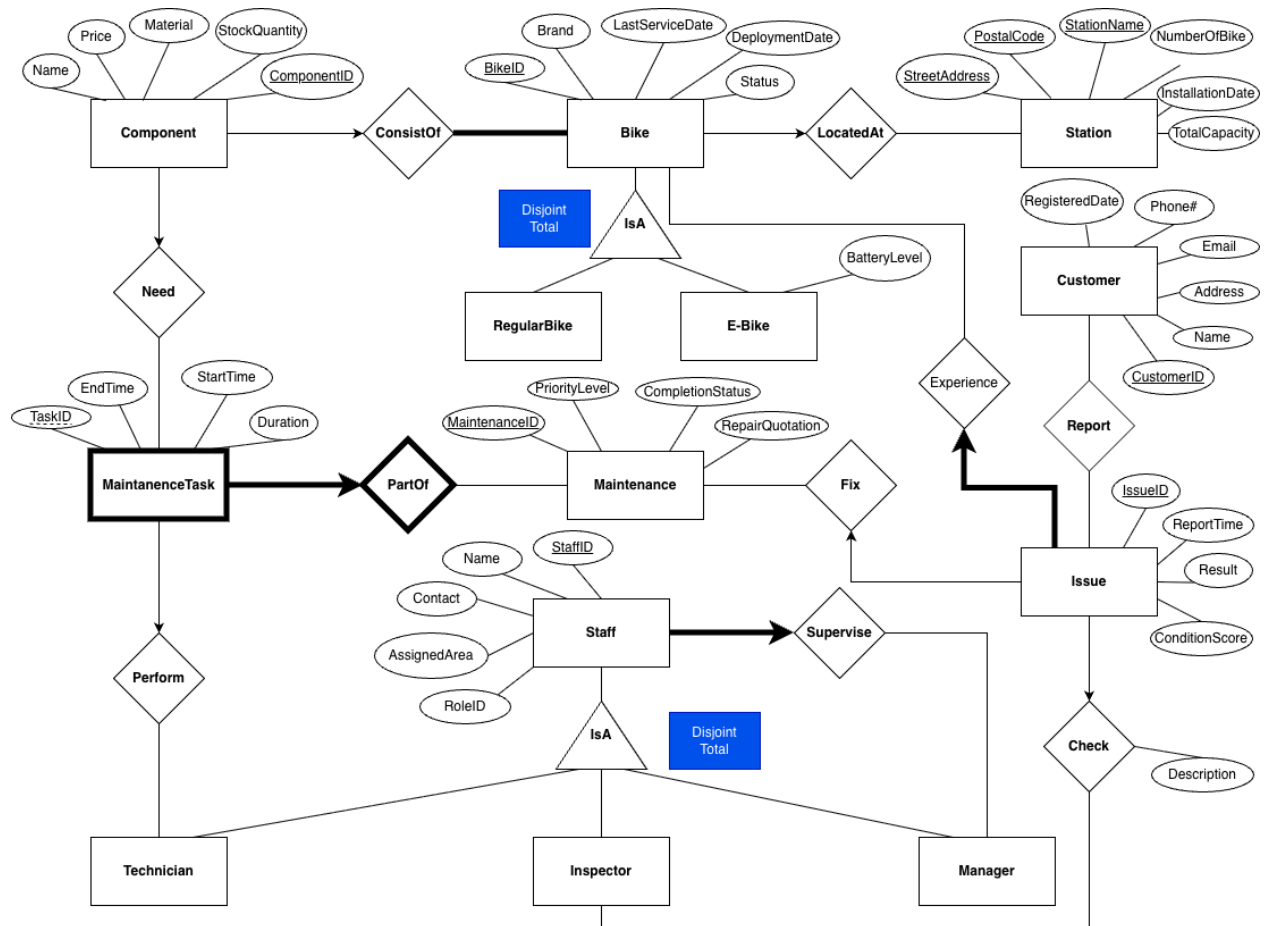
Cardinality:

- Inspector → Issue: One-to-Many

Constraint:

- Key constraint on Issue

ER Diagram



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Additional Comments

During the initial brainstorming phase, our team considered designing a general bicycle maintenance database that would support both privately owned bicycles and shared urban bikes. However, as we discussed the data modeling implications, we identified challenges associated with uniquely identifying privately owned bicycles. Private bikes vary widely in brand, model, and ownership, and there is no standardized or reliable identifier that could consistently serve as a primary key across different users and contexts.

In contrast, shared urban bike systems provide a well-defined structure in which bicycles are owned and managed by a single organization. Each bicycle is assigned a system-generated identifier, making it natural and reliable to uniquely identify bikes using a primary key such as “BikeID”. This significantly simplifies entity identification, relationship modeling, and maintenance tracking within the database.

Based on these considerations, we decided to focus our application on urban bike infrastructure maintenance. This decision allowed us to build a more coherent and well-structured database that connects bike deployment, issue reporting, maintenance workflows, and staff responsibilities. As a result of this focus, the role of customers in the system is more limited: customers primarily interact with the system by reporting issues encountered during bike usage, rather than being owners of the bicycles themselves. We believe this scope choice leads to a clearer and more realistic representation of real-world shared bike maintenance operations.

Acknowledgment of Using AI Tools

Yes. AI tools (e.g., ChatGPT) were used for language polishing, clarification of written explanations, and word choice refinement. The ER design, domain analysis, and all technical content were independently produced by the project team.